M1 Understanding probability

Name:

Class:

Date:

Time: **330 minutes**

Marks: **266 marks**

Comments:

Q1.

(a) A cube has faces labelled 1, 1, 2, 2, 3 and 3. The cube is to be rolled once, and each face is equally likely to appear on top.

Write down the probability that the number showing will be 1.

(1)

(b) Two cubes, both identical to the cube described in part (a), are each to be rolled once.

Find the probability that:

(i) both of the numbers showing will be 1;

(2)

(ii) neither of the numbers showing will be 1.

(2)
(Total 5 marks)

Q2.

Eileen is the secretary of a book club. She wishes to purchase three copies of a particular novel for members to read, prior to discussion at the next meeting. There are two book shops in the town: Sherrats and Hughes. The table below shows the number of copies of the novel in stock, together with their associated probabilities at each shop. The probabilities at each shop are independent of those at the other shop.

Number of copies in stock	**Probability**	
	Sherrats	**Hughes**
0	0.15	0.30
1	0.25	0.28
2	0.20	0.18
3 or more	0.40	0.24

(a) Find the probability that:

(i) neither book shop has any copies in stock;

(1)

(ii) Sherrats has exactly one copy in stock and Hughes has two or more copies in stock;

(3)

(iii) the two book shops have, between them, a total of 3 or more copies in stock.

(4)

(b) Eileen decides to visit Sherrats and buy as many copies as possible, up to a maximum of 3. If she is unable to buy 3 copies at Sherrats, she will then visit Hughes. At Hughes, she will buy sufficient copies to make her total up to 3 or, if this is not possible, she will buy as many copies as are in stock.

Find the probability that Eileen buys:

(i) three copies from the same shop;

(3)

(ii) more copies from Hughes than from Sherrats.

(4)

(Total 15 marks)

Q3.

Henry passes through 6 sets of traffic lights when he drives to work each morning. He estimates that the probability of each set of traffic lights showing green as he approaches is 0.3.

Assume that the probabilities are independent. As he drives to work one morning, find the probability that as he approaches:

(a) 2 or fewer sets show green;

(b) more than 3 sets show green;

(c) all 6 sets show green.

(Total 7 marks)

Q4.

Veronica reviews films for a weekly magazine. She awards each film one, two, three or four stars according to how much she enjoys it. She classifies films as *comedy, drama* or *other.* The following table shows the probability that Veronica awards a given number of stars to each class of film.

	One star	**Two star**	**Three star**	**Four star**
Comedy	0.25	0.45	0.24	0.06
Drama	0.20	0.35	0.30	0.15
Other	0.40	0.10	0.32	0.18

(a) Find the probability that Veronica will award more than two stars to a *comedy.*

(1)

During a particular six-week period, Veronica reviews 30 films: 10 *comedies,* 15 *dramas* and 5 *others.*

(b) One of these 30 films is selected at random. Find the probability that it is:

(i) a *comedy;*

(1)

(ii) a *comedy* awarded four stars by Veronica;

(1)

(iii) a *drama* awarded more than two stars by Veronica;

(2)

(iv) awarded four stars by Veronica;

(3)

(v) awarded fewer than three stars by Veronica, given that it is **not** a *comedy*.

(3)

(c) If three of the 30 films are selected at random without replacement, find the probability that two are *comedies* and one is a *drama*.

(3)
(Total 14 marks)

Q5.

In a target golf contest, Anne, Brian and Gurwyn each hit a ball towards a ring. The probability that a ball finishes inside the ring is $\frac{3}{4}$ for Anne, $\frac{2}{3}$ for Brian and $\frac{4}{5}$ for Gurwyn.

These probabilities may be considered to be independent. Find the probability that:

(a) none of the three balls finishes inside the ring;

(2)

(b) at least one of the three balls finishes inside the ring;

(1)

(c) exactly two of the three balls finish inside the ring;

(3)

(d) the ball hit by Gurwyn does not finish inside the ring, given that exactly two of the three balls finish inside the ring.

(2)
(Total 8 marks)

Q6.

Maurice works at home. At 2 pm he decides to take a break to buy a copy of the Chronicle newspaper. There are three nearby newsagents: Arif, Bob and Carol. However by 2 pm they may have sold all their Chronicles and so have none available. The independent probabilities that they have a Chronicle available at 2 pm are:

Arif 0.4 Bob 0.7 Carol 0.25

(a) State the probability that Bob does not have a Chronicle available at 2 pm.

(1)

(b) Find the probability that none of the three newsagents has a Chronicle available at 2 pm.

(3)

(c) (i) Find the probability that Bob does not have a Chronicle available at 2 pm but Arif does.

(ii) Find the probability that Carol does not have a Chronicle available at 2 pm but Arif does.

(3)

(d) Maurice decides to visit the newsagents in turn until he obtains a Chronicle or until he has visited all three. He tosses a coin. If it lands heads he will visit the three newsagents in the order Bob, Arif, Carol. If it lands tails he will visit them in the order Carol, Arif, Bob.

Find the probability that he will obtain a Chronicle from Arif.

(3)
(Total 10 marks)

Q7.

On a visit to a theme park, the probabilities that a visitor will go on the Monorail, the Big Mountain ride and the Fearsome ride are 0.65, 0.52 and 0.38 respectively.

The probability that a visitor will go on the Big Mountain ride and the Fearsome ride is 0.25.

The probability that a visitor will go on the Monorail and the Big Mountain ride is 0.30. The probability that a visitor will go on the Fearsome ride, given that they go on the Monorail is 0.46.

Find the probability that a randomly selected visitor to this theme park will go on:

(a) the Monorail and the Fearsome ride;

(2)

(b) the Monorail or the Big Mountain ride or both;

(2)

(c) neither the Big Mountain ride nor the Fearsome ride;

(3)

(d) the Monorail, given that they go on the Fearsome ride.

(3)
(Total 10 marks)

Q8.

A petrol station sells three types of fuel: lead replacement petrol (LRP), unleaded petrol and diesel. The probability that a randomly selected customer will buy a particular fuel is shown in the table.

Fuel	Probability
LRP	0.15
Unleaded	0.65
Diesel	0.20

(a) Find the probability that of two randomly selected customers:

(i) both will buy diesel;

(ii) exactly one will buy diesel.

(5)

(b) Find the probability that of three randomly selected customers:

(i) all will buy unleaded;

(ii) exactly two will buy diesel;

(iii) one will buy LRP, one will buy unleaded and one will buy diesel.

(7)
(Total 12 marks)

Q9.

The age and the blood pressure status for each adult male in a randomly selected sample are summarised in the following table.

Blood pressure status \ Age Range	25 – 34	35 – 54	55 – 74
Normal (untreated)	52	47	28
High (untreated)	4	7	18
High (treated)	1	4	13

One of the adult males is selected at random.

H is the event ‘the male selected has high blood pressure’.
R is the event ‘the male selected is aged 55–74’.
S is the event ‘the male selected is aged 25–34’.
T is the event ‘the male selected is being treated’.

R′ is the event ‘not R’.
H′ is the event ‘not H’.

(a) Find:

(i) P(S);

(1)

(ii) P(T);

(1)

(iii) $P(R \cup T)$;

(2)

(iv) $P(H \cap R')$;

(2)

(v) $P(S|T)$;

(3)

(vi) $P(R|H')$.

(3)

(b) Find the probability of the event $(R \cap H \cap T)$.

Define this event in context as simply as possible.

(3)
(Total 15 marks)

Q10.

When Gordon and Louise play a board game, the probability that Gordon wins is 0.6 and the probability that Louise wins is 0.4. They agree to play a series of games. The winner of the series will be the first player to win three games. The result of each game is independent of the result of any other game.

Find the probability that:

(a) Gordon wins the series by three games to nil;

(2)

(b) more than three games are necessary to decide who wins the series;

(3)

(c) Louise wins the series by three games to one, given that she wins the first game;

(3)

(d) Louise wins the series, given that the final score is three games to two;

(2)

(e) Gordon wins the series, given that Louise wins the first game.

(4)
(Total 14 marks)

Q11.

A company analyses the servicing of machinery at one of its large factories.

When a machine is serviced:

B is the event 'the bearings need replacement';
C is the event 'the compressor pump needs replacement';
S is the event 'the solenoid needs replacement'.

The events B, C and S are independent.

B' is the event 'not B'.
C' is the event 'not C'.

A machine is selected at random for servicing.

You are given that P(B) = 0.1 and P(B $\cup$ C) = 0.28.

(a) (i) Show that P(C) = 0.2.

(4)

(ii) Find the value of P(C' $\cup$ B').

(2)

(b) The probability that the solenoid will need replacement is 0.3.

Find the probability that:

(i) the bearings will need replacement, but neither the compressor pump nor the solenoid will need replacement;

(2)

(ii) exactly **one** of the three components will need replacement;

(3)

(iii) the solenoid will need replacement, given that exactly one of the three components needs replacement.

(3)

(Total 14 marks)

Q12.

George and Jose are friends who play together after school. On fine days, they have a choice of playing computer games or going to the park. The probability that, on any fine day, they will prefer to play computer games is 0.8 for George and, independently, 0.7 for Jose.

(a) Find the probability that on a particular fine day:

(i) both George and Jose prefer to play computer games;

(ii) both George and Jose prefer to go to the park;

(iii) exactly one of George or Jose prefers to go to the park.

(5)

(b) Tony is another friend who plays with George and Jose after school. The probability that, on a fine day, Tony prefers to play computer games is 0.95 when George prefers to play computer games and 0.15 when George prefers to go to the park. Tony's preferences are independent of those of Jose.

Find the probability that on a particular fine day:

(i) all three boys prefer to play computer games;

(ii) two or more of the boys prefer to play computer games.

(5)
(Total 10 marks)

Q13.

An airline company accepts bookings for 20 executive-class seats on a flight from London to Bergen. The probability that a customer, who has booked an executive-class seat, will not turn up to claim the seat is 0.15 and may be assumed to be independent of the behaviour of other customers.

(a) Find the probability that, of the customers who have booked executive-class seats:

(i) all will turn up to claim their seats;

(ii) one or more will **not** turn up to claim their seat.

(4)

(b) The airline also accepts bookings for 50 standard-class seats on the same flight. The probability that a customer, who has booked a standard-class seat, will not turn up to claim the seat is 0.08 and may be assumed to be independent of the behaviour of other customers.

Find the probability that, of the customers who have booked standard-class seats:

(i) 2 or more will **not** turn up;

(ii) 3 or more will **not** turn up

(4)

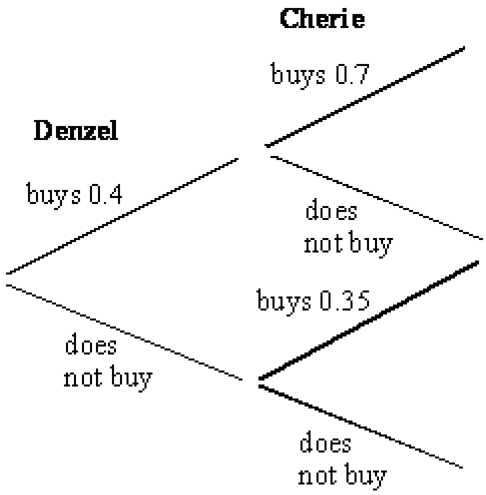
(c) The aeroplane has 19 executive-class seats and 48 standard-class seats. If 19 or fewer customers who have booked executive-class seats turn up they will all be allocated executive-class seats. If all 20 customers who have booked executive-class seats turn up, one will be allocated a standard-class seat. This will leave only 47 seats available for the 50 customers who have booked standard-class seats.

Use your results from parts (a) and (b) to calculate the probability that there will be standard-class seats available for all those who turn up having booked standard-class seats.

(5)
(Total 13 marks)

Q14.

Denzel and Cherie are friends who often go to the cinema together. On such visits there is a probability of 0.4 that Denzel will buy popcorn. The probability that Cherie will buy popcorn is 0.7 if Denzel buys popcorn and 0.35 if he does not.



(a) When Denzel and Cherie visit the cinema together:

(i) find the probability that both buy popcorn;

(2)

(ii) show that the probability that neither buys popcorn is 0.39;

(2)

(iii) find the probability that exactly one of them buys popcorn.

(2)

(b) Sohaib sometimes joins Denzel and Cherie on their cinema visits. On these occasions, the probability that Sohaib buys popcorn is 0.55 if both Denzel and Cherie buy popcorn and 0.25 if exactly one of Denzel and Cherie buys popcorn.

Find the probability that when Denzel, Cherie and Sohaib visit the cinema together:

(i) all three buy popcorn;

(2)

(ii) Cherie and Sohaib buy popcorn but Denzel does not.

(2)

(Total 10 marks)

Q15.

Jamal is one of 10 employees of a children's charity. The charity shares a small office building with two other charities, each with 6 employees. It is agreed that 3 of these 22 employees will be chosen by a random process to form a building management committee.

Find the probability that:

(a) Jamal is chosen for the committee;

(1)

(b) all three members of the committee are employed by Jamal's charity;

(3)

(c) all three members of the committee are employed by the same charity;

(4)

(d) exactly two employees of Jamal's charity are chosen for the committee;

(4)

(e) Jamal and exactly one other employee of his charity are chosen for the committee.

(4)

(Total 16 marks)

Q16.

A rugby club has three categories of membership: adult, social and junior. The number of members in each category, classified by gender, is shown in the table below.

	Adult	**Social**	**Junior**
Female	25	35	40
Male	95	25	80

One member is chosen, at random, to cut the ribbon at the opening of the new clubhouse.

(a) Find the probability that:

(i) a female member is chosen;

(ii) a junior member is chosen;

(iii) a junior member is chosen, given that a female member is chosen.

(4)

(b) V denotes the event that a female member is chosen.
W denotes the event that an adult member is chosen.
X denotes the event that a junior member is chosen.

For the events V, W and X:

(i) write down two which are mutually exclusive;

(1)

(ii) find two which are neither mutually exclusive nor independent. Justify your answer.

(3)

(Total 8 marks)

Q17.

Ricky, Saeed and Tom like to go to the park after school on Wednesday to play football.

The independent probabilities that Ricky, Saeed and Tom go to the park to play football on Wednesday are 0.8, 0.7 and 0.9 respectively.

Find the probability that, on a particular Wednesday:

(a) none of the three boys goes to the park;

(2)

(b) Ricky is the only one to go to the park;

(2)

(c) only **one** of the three boys goes to the park;

(3)

(d) Ricky goes, given that only one of the three boys goes to the park.

(3)
(Total 10 marks)

Q18.

Transport inspectors carry out roadside safety tests on lorries.

(a) The probability of a randomly selected lorry failing the test is 0.25. A transport inspector chooses two lorries at random. Find the probability that:

(i) both will fail the test;

(ii) exactly one will pass the test.

(5)

(b) Of 12 lorries parked outside a transport café, four would fail if tested.

(i) An inspector chooses two of these twelve lorries at random. Find the probability that both will pass the test.

(2)

(ii) An inspector chooses three of these lorries at random. Find the probability that two will pass the test and one will fail the test.

(3)

(iii) An inspector chooses four of these lorries at random. Find the probability that at least one will fail the test.

(3)
(Total 13 marks)

Q19.

Shahid, Tracy and Dwight are friends who all have birthdays during January. Assuming that each friend's birthday is equally likely to be on any one of the 31 days of January, find the probability that:

(a) Shahid's birthday is on January 3rd;

(1)

(b) both Shahid's and Tracy's birthdays are on January 3rd;

(2)

(c) all three friends' birthdays are on the same day;

(2)

(d) all three friends' birthdays are on different days.

(3)
(Total 8 marks)

Q20.

Amy, Bruce and Carmen take part in an archery competition. The final test of the competition involves these three archers trying to hit the target centre.

The independent probabilities that Amy, Bruce and Carmen hit the target centre are 0.2, 0.3 and 0.6 respectively.

Find the probability that the target centre will be hit by:

(a) Bruce only;

(2)

(b) Carmen only;

(2)

(c) exactly one of the three archers;

(3)

(d) Bruce, given that exactly one of the three archers hits the target centre.

(3)
(Total 10 marks)

Q21.

Jennifer attends college five days a week. Each day she is either on time or late. The following table shows the probability that she is late for each day that she attends college.

Day	**Monday**	**Tuesday**	**Wednesday**	**Thursday**	**Friday**
Probability late	0.4	0.1	0.1	0.1	0.55

(a) Find the probability that, in a particular week, Jennifer is:

(i) on time every day;

(ii) late on at least one day.

(4)

(b) The probability that Jennifer's friend Hassan is late for college is 0.7 on days when Jennifer is late and 0.05 on days when Jennifer is on time.

Find the probability that, on a particular Monday:

(i) both Jennifer and Hassan are on time;

(ii) one is on time and the other is late.

(5)
(Total 9 marks)

Q22.

A blood test for determining the type of hepatitis from which a patient is suffering involves testing for the Australian antigen. A "positive" result to this test indicates that a patient has hepatitis C.

Unfortunately, this test is not completely reliable.

If an individual has hepatitis C, there is a probability of 0.8 that the test will give a "positive" result. If a patient does **not** have hepatitis C, there is a probability of 0.1 that the test will give a "positive" result.

It is known that 18% of patients suffering from hepatitis have type C.

Find the probability that a person chosen at random from all hepatitis sufferers:

(a) has type C and gives a "positive" result to the test;

(2)

(b) gives a "positive" result to the test;

(3)

(c) does not have hepatitis C, given that the test has given a "positive" result.

(3)
(Total 8 marks)

Q23.

Following a flood, 120 tins were recovered from Dharmesh's corner shop. Unfortunately the water had washed off all the labels. Of the tins, 50 contained pet food, 20 contained peas, 35 contained beans and the rest contained soup.

(a) Dharmesh selects a tin at random. Find the probability that it:

(i) contains soup;

(1)

(ii) does not contain pet food.

(1)

(b) Dharmesh selects two tins at random (without replacement). Find the probability that:

(i) both contain peas;

(2)

(ii) one contains pet food and the other contains peas.

(3)

(c) Dharmesh selects three tins at random (without replacement). Find the probability that one contains pet food, one contains peas and one contains beans.

(3)

(d) Find the probability that Dharmesh will have to open more than two tins before she finds one which does not contain pet food.

(3)

(Total 13 marks)

Q24.

An open prison purchases a large number of tins of baked beans which are supplied by three wholesalers: Yeadon, Khan and Heinrich.

25% of the tins are supplied by Yeadon of which 5% are underweight.

30% of the tins are supplied by Khan of which 2% are underweight.

45% of the tins are supplied by Heinrich of which 3% are underweight.

(a) Find the probability that a randomly selected tin:

(i) has been supplied by Yeadon and is underweight;

(2)

(ii) is underweight.

(4)

Of those tins supplied by Yeadon, 0.9% are both underweight and have lost their label.

Of those **underweight** tins supplied by Khan, 20% have lost their label.

Of those tins supplied by Heinrich, 25% have lost their label. The probability of Heinrich's tins having lost their label is independent of whether or not they are underweight.

(b) Find the probability that a tin is both underweight and has lost its label if it has been selected at random from the tins supplied by:

(i) Yeadon;

(ii) Khan;

(iii) Heinrich.

(5)

(c) Find the probability that a randomly selected tin is both underweight and has lost its label.

(3)

(Total 14 marks)